calculations into a perturbative framework based on the expectation that a reasonable ballpark estimate is given by the zero-loop processes, with the loop diagrams resulting in refinements that get smaller as the number of loops increases.

In fact, almost everything we know about string theory—including much of the material covered in previous chapters—has been discovered by physicists performing detailed and elaborate calculations using this perturbative approach. But to trust the accuracy of the results found, one must determine whether the supposedly ballpark approximations that ignore all but the first few diagrams in Figure 12.6 are really in the ballpark. This leads us to ask the crucial question: Are we in the ballpark?

Is the Ballpark in the Ballpark?

It depends. Although the mathematical formula associated with each diagram becomes very complicated as the number of loops grows, string theorists have recognized one basic and essential feature. Somewhat as the strength of a rope determines the likelihood that vigorous pulling and shaking will cause it to tear into two pieces, there is a number that determines the likelihood that quantum fluctuations will cause a single string to split into two strings, momentarily yielding a virtual pair. This number is known as the string coupling constant (more precisely, each of the five string theories has its own string coupling constant, as we will discuss shortly). The name is quite descriptive: The size of the string coupling constant describes how strongly the quantum jitters of three strings (the initial loop and the two virtual loops into which it splits) are related—how tightly, so to speak, they are coupled to one another. The calculational formalism shows that the larger the string coupling constant, the more likely it is that quantum jitters will cause an initial string to split apart (and subsequently rejoin); the smaller the string coupling constant, the less likely it is for such virtual strings to erupt momentarily into existence.

We will shortly take up the question of determining the value of the string coupling constant within any of the five string theories, but first, what do we really mean by “small” or “large” when assessing its size? Well, the mathematics underlying string theory shows that the dividing line between “small” and “large” is the number 1, in the following sense. If the string coupling constant has a value less than 1, then—